

1.

```
class Node {  
    int data;  
    Node next;  
  
    //constructor  
  
    Node(int d){  
        data = d;  
        next = null;  
    }  
}
```

2.

```
class LinkedList {  
  
    Node head; // head of the list  
    Node tail; // tail of the list  
  
    // creating an empty linked list  
    LinkedList() {  
        this.head = null;  
        this.tail = null;  
    }  
  
    // Insert Front  
  
    public void insertFront(int new_data) {  
  
        Node new_node = new Node(new_data);  
  
        if (head == null) {  
            head = new_node;  
            tail = new_node;  
        } else {  
            new_node.next = head;  
            head = new_node;  
        }  
    }  
}
```

// Insert Rear

```
public void insertRear(int new_data) {
    Node new_node = new Node(new_data);

    if (head == null) {
        head = new_node;
        tail = new_node;
    } else {
        tail.next = new_node;
        tail = new_node;
    }
}
```

// Search

```
public Node search(Node head, int x) {
    Node current = head;
    if (nodeExists(current, x) == true) {
        while (current != null) {
            if (current.data == x) {
                return current;
            }
            current = current.next;
        }
    } else {
        return null;
    }
    return null;
}
```

// Display

```
public void display() {
    Node temp = head;

    while (temp != null) {
        System.out.print(temp.data + " ");

        temp = temp.next;
    }
    System.out.println();
}
```

3.

```
class Driver {  
  
    public static void main(String args[]) {  
  
        LinkedList list1 = new LinkedList();  
  
//a        list1.insertRear(20);  
           list1.insertRear(30);  
           list1.insertRear(40);  
           list1.insertRear(50);  
  
//b        //list1.display();  
  
           list1.insertFront(10);  
  
//c        //list1.display();  
  
           list1.insertRear(60);  
  
    }  
  
}
```